

ROCKY MOUNTAIN HOUSE, AB, Canada

WMO: 719280

Lat: 52.4214N

Lon: 114.9122W

Elev: 3242

StdP: 13.05

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-19.3	-34.1	0.9	-25.5	-26.8	1.4	-18.4	20.6	31.6	17.9	27.7	2.4	120	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	80.1	61.5	76.9	59.9	73.9	58.7	64.1	76.0	62.1	73.8	60.2	70.9	7.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	85.7	70.0	57.4	79.2	67.1	55.5	74.1	64.7	31.0	76.0	29.4	74.2	28.0	71.2	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.2	13.1	DB	-35.9	86.1	6.8	3.1	-40.7	88.4	-44.7	90.2	-48.5	92.0	-53.4	94.2	(4)
(5)			WB	-35.4	67.7	5.8	2.3	-39.5	69.4	-42.9	70.7	-46.2	72.0	-50.4	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	12.1	14.8	23.2	36.5	47.0	53.9	59.2	57.0	48.7	37.5	23.0	12.7		
	DBStd	19.94	15.46	13.72	13.65	8.96	7.47	5.27	5.23	5.62	6.96	8.19	12.36	13.86		
	HDD50	6038	1175	984	832	409	146	20	2	8	105	392	810	1155		
	HDD65	10754	1640	1404	1297	855	558	335	191	252	489	853	1260	1620		
	CDD50	775	0	0	0	4	53	136	286	226	66	4	0	0		
	CDD65	15	0	0	0	0	0	1	10	4	0	0	0	0		
	CDH74	641	0	0	0	1	46	63	273	201	55	2	0	0		
	CDH80	100	0	0	0	0	4	9	48	32	7	0	0	0		
	WSAvg	4.9	4.3	4.5	5.5	6.2	6.1	5.7	5.2	4.4	4.4	4.7	4.1	4.1		
	PrecAvg	21.3	0.9	0.7	0.9	1.2	2.5	4.2	3.5	2.9	1.8	0.9	0.8	0.7		
(15) Precipitation	PrecMax	25.5	2.3	1.5	1.6	3.5	4.3	7.6	8.1	5.4	3.9	2.2	1.4	1.7		
(16)	PrecMin	15.9	0.2	0.2	0.2	0.3	0.9	0.9	0.7	0.8	0.2	0.2	0.2	0.2		
(17)	PrecStd	3.1	0.4	0.4	0.4	0.8	1.0	1.7	1.4	1.1	1.1	0.6	0.4	0.5		
(18)																
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	50.1	55.3	70.3	79.2	79.9	84.6	83.6	79.9	70.3	57.1	48.3		
		MCWB	38.3	37.8	40.9	50.0	55.4	60.4	63.3	62.6	58.5	50.9	44.3	37.6		
	2%	DB	44.9	43.7	49.1	63.8	74.0	75.1	80.5	79.1	74.5	62.9	49.7	42.1		
		MCWB	35.5	34.1	37.8	47.0	53.2	58.0	63.5	62.0	57.0	48.3	39.4	33.7		
	5%	DB	39.3	38.6	44.9	58.4	69.2	71.6	76.8	75.6	70.0	57.4	43.9	37.1		
		MCWB	32.2	31.1	35.5	43.7	51.5	56.3	61.6	60.6	55.4	45.3	35.7	30.6		
	10%	DB	33.1	34.1	41.1	52.7	64.5	68.0	73.3	72.0	65.0	52.3	38.9	31.9		
		MCWB	29.0	28.5	33.4	40.8	49.2	54.5	60.0	59.1	53.0	42.5	33.2	27.8		
	0.4%	WB	38.7	38.4	42.3	50.4	57.4	62.8	67.7	66.1	60.9	53.1	44.9	37.9		
		MCDB	49.1	49.1	53.5	67.5	74.8	76.6	79.0	77.9	74.9	67.8	56.0	47.1		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	34.7	38.5	47.6	54.7	60.0	65.0	63.5	58.4	49.0	40.1	34.2		
(29)		MCDB	44.9	42.7	47.7	62.3	70.5	71.2	76.8	75.6	72.0	61.1	48.5	41.6		
(30)	5%	WB	32.7	31.5	36.0	44.7	52.4	57.9	62.9	61.6	56.1	46.0	36.6	30.9		
(31)		MCDB	38.6	37.5	44.1	57.2	66.6	68.8	74.6	73.2	68.1	56.1	42.7	36.4		
(32)	10%	WB	29.0	28.9	33.7	41.5	50.0	55.8	60.9	59.8	53.7	43.1	33.4	27.7		
(33)		MCDB	33.5	33.8	40.5	51.5	62.6	65.7	71.3	70.6	64.2	51.0	38.0	31.8		
(34)																
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	24.2	21.7	22.5	24.6	22.5	23.5	24.6	25.3	23.4	20.8	21.4		
(36)		MCDBR	28.3	29.6	27.5	34.2	34.9	30.8	30.5	32.4	36.7	33.5	27.3	26.0		
(37)		MCWBR	20.1	20.9	17.4	19.2	17.5	15.7	15.6	17.0	20.7	19.7	17.8	18.8		
(38)		MCDBR	27.3	28.2	25.8	32.3	31.7	26.8	27.8	29.0	33.6	31.8	25.8	25.4		
(39)	5% WB	MCWBR	20.0	20.3	16.8	18.5	16.5	14.4	15.1	15.9	19.3	19.2	17.5	18.6		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.261	0.296	0.331	0.344	0.353	0.350	0.360	0.318	0.283	0.255	0.242		
(41)	taud		2.426	2.486	2.447	2.381	2.395	2.389	2.434	2.384	2.485	2.578	2.521	2.411		
(42)	Ebn at Noon		246	280	287	287	288	285	284	274	276	268	247	231		
(43)	Edh at Noon		17	22	28	34	35	36	34	34	27	20	16	15		
(44) All-Sky Solar Radiation	RadAvg		321	613	1052	1452	1711	1814	1935	1611	1131	682	360	248		
(45)	RadStd		19	40	62	115	121	137	121	106	123	71	28	17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page